

Department of Electrical and Information Engineering

University of Ruhuna

Mini Project Proposal

Image Processing and Computer Vision

EC7205

Restricted Area Monitoring and Unauthorized Entry Detection System

EG/2021/4475 - Dewthilini G.W.B.S

EG/2021/4468 - De Silva W.D.L.

EG/2021/4418 - Athapaththu A.A.V.S.

EG/2021/4524 - Gunarathna B.V.I.D.W

Jan/26/2026

Department of Electrical and Information Engineering
University of Ruhuna

Contents

Contents 1

1 Problem Statement 2

2 Objectives 3

3 Literature Review 4

4 Methodology 8

4.1 Input and Preprocessing . . . 8

4.1.1 Video Input Acquisition . . . 8

4.1.2 Frame Extraction . . . 8

4.1.3 Frame Resizing and Normalization . . . 9

4.1.4 Noise Reduction . . . 9

4.1.5 Noise Reduction . . . 9

4.1.6 Lighting and Contrast Enhancement. . . . 9

4.2 Restricted Area Detection . . . 10

4.2.1 Region of Interest (ROI) Definition . . . 10

4.2.2 ROI Mask Generation . . . 10

4.2.3 Object Presence Detection and Motion. . . . 10

4.2.4 Rejection of irrelevant Movements . . . 10

4.3 Object Detection and Classification . . . 11

4.3.1 Object Detection Model Selection . . . 11

4.3.2 Detection Process within Restricted Area . . . 11

4.3.3 Multi-Class Object Classification . . . 11

4.3.4 False Object and Noise Handling . . . 12

4.4 Specialized Identification and Authentication . . . 12

4.4.1 Human Detection and Face Analysis . . . 12

4.4.2 Vehicle Detection and Number Plate Recognition . . . 12

4.4.3 Animal Detection and Classification . . . 13

4.4.4 Data Storage and Logging . . . 13

4.5 Decision Making and Alert Generation . . . 13

5 Dataset / Data Collection 14

6 Expected Outcomes 16

7 Percentage of Image Processing & Computer Vision Involvement 17

1. Problem Statement

Areas such as construction sites, industry zones, wildlife habitats, and restricted areas need surveillance to restrict the access of unwanted individuals to their premises. Current surveillance technologies are mostly applicable for the monitoring of general movements or whole-area surveillance. But most surveillance technologies are not effectively applied to these matters, as they are mostly applicable to the monitoring of human movements and are not applicable to the monitoring of movements by animals or vehicles, which also pose risks to the concerned premises. Moreover, monitoring through CCTV cameras is time-consuming and also vulnerable to human error.

This project proposes to overcome all these limitations by developing a system called "Restricted Area Monitoring and Unauthorized Entry Detection" using computer vision and image processing techniques. This proposed system works only in restricted regions and accurately detects unauthorized activities. It also captures all activities that take place in those regions efficiently.It accurately detects activities taking place unauthorized access by both humans and animals, and also vehicles. In the context of security, the suggested system will enhance security through automation in detecting and alert generation.

Back to Table of Contents

2. Objectives

- **Capture and preprocess video input**

  To continuously take a video feed from security cameras, as well as to increase the quality of each frame by employing a few image processing operations likeresizing, filtering out noise, and enhancing colors, so that objects can be detected in various lighting/weather conditions.

- **Define and monitor restricted areas**

  To identify and monitor certain restricted areas of interest in the camera's line of sight. By monitoring these areas of interest only, there will be less processing of information in the scene and less inaccuracies from movements outside this restricted zone of interest.

- **Detect humans, animals, and vehicles**

  To identify all kinds of entities that are intruding into the restricted area by making use of computer vision technology. So, in addition to human, entities such as animal and vehicle are also taken into consideration.

- **Identify unauthorized entry**

  To identify if the identified entities passing into the restricted zone are authorized to be in the specific region of interest, the system has an evaluation mechanism based on the defined rules or an authorized database on the nature of the identified entities passing into the restricted region. For humans, this can be performed through identity recognition technologies such as face recognition systems. For vehicles, the system may compare vehicle type or license plate information with a list of approved entries. For animals, the system checks to identify if the identified animals are allowed in the region of interest or regarded as an intrusion. These checks assure the system that only authorized entities are passing into the restricted region while any intruders are easily identified.

Back to Table of Contents

3. Literature Review

Video-based security systems are now frequently used to protect residential and private properties due to the rise of smart homes and affordable priced security cameras. However, motion detection is the primary method used by traditional surveillance systems, which frequently results in a large number of false alarms. Using computer vision and deep learning techniques like object detection, face recognition, and vehicle identification, recent research has concentrated on enhancing surveillance systems. This section reviews a number of recent studies on human identification, automatic number plate recognition, region-based monitoring, and unauthorized entry detection. To determine the research gap that the proposed system fills, the advantages and disadvantages of these works are reviewed.

- **Study 1: Motion-Based Surveillance Using Background Subtraction.**

  Older video surveillance systems typically used basic motion detection techniques such as background subtraction and Gaussian Mixture Models (GMM) to identify moving objects by comparing video frames to identify any changes captured by the camera. The research paper "Moving Object Detection in Visual Surveillance Based on Background Subtraction" [1] describes how basic background subtraction was used to identify moving objects within a video sequence taken from a static camera. According to this paper, their background subtraction work for detecting moving video objects works well in simple indoor environments and it can be executed on lower powered computers. However, basic motion detection systems produce numerous false alarms due to various factors such as shadowing, changes in light, animals and vehicle traffic. Also, basic motion detection systems cannot determine exactly what type or classification of object is moving. Because of these limitations, motion detection systems based upon basic motion detection will not provide accurate or reliable outputs for monitoring real-world residential areas that are within major traffic areas. The other major limitation of prior motion detection systems is that all are designed to treat every motion as being the same and do not account for whether or not there is a legitimate reason for that motion (eg humans, animal, vehicle, etc). To resolve this problem, we hope to develop an improved video surveillance system that uses an object based method of identifying moving objects instead of using basic motion detection across an area that has been defined and restricted to provide increased accuracy and access to reduced false alarm frequency.

- **Study 2: Object Detection-Based Surveillance Using YOLO.**

  Many current surveillance systems rely on deep learning to find and label objects in videos, and YOLO (You Only Look Once) is one of the models most often used for this task. The model identifies and labels objects such as people, vehicles, and ani-

mals in live videos with low delay, which supports real-time monitoring tasks. One example is the paper Real-Time Moving Object Detection Using YOLO [2]. In that study, the authors modified YOLO so it could detect moving objects quickly while still keeping good accuracy, which they aimed to apply to surveillance settings. There is another related article "Real Time Object Detection based on YOLOv3 using Python" [3], which gives a practical, step-by-step YOLOv3 implementation for surveillance video analysis in Python. In these systems, YOLO uses a convolutional neural network to identify the object types in an image and predict their locations within the frame. A key advantage of YOLO is that it can identify multiple object classes in a single pass while keeping the speed needed for real-time work in outdoor settings. One limitation is that it detects objects across the entire frame, including areas that are irrelevant to security. It cannot determine whether a detected person or vehicle has permission to be in the area, and it cannot identify specific people or vehicles. As a result, earlier YOLO-based studies in surveillance often still have trouble distinguishing routine behavior from suspicious activity. We hope to address this issue by combining YOLO with a predefined Region of Interest (ROI) and identity verification steps, which reduces false alerts and keeps the detection focused on relevant areas.

- **Study 3: ROI-Based Object Detection Optimization**

  Researchers have done studies to improve Object Detection by focusing on important areas in an image (also known as Region Of Interest or ROI). With ROI methods, instead of processing every single pixel in the video frames, the background noise is first removed and all object detection will happen in the areas of the image that really matter. Jeon and Lee (2025) published an article called "Region of Interest Extraction Method to Increase Object Detection Performance in Remote Monitoring System" [4], which discusses their methods for extracting meaningful areas prior to performing object detection to reduce processing costs and increase efficiency. To perform object detection using their method, they applied ROI masking prior to object detection; that is, they will only process the portions of every frame that have the potential to contain objects rather than analysing the entire frame. By reducing the amount of time or energy spent analysing background areas of a frame the performance gains in reducing false detections and speeding up processing are achieved. The most significant benefits of This method are the reduction of the computational load required and the increase in speed of detection, which is especially advantageous when working with limited hardware or remote monitoring systems.

- **Study 4: Face Recognition-Based Surveillance Systems**

  Face recognition has been widely studied as a method for identify people in surveillance systems. There have been many researchers using face recognition technology

to make security systems smarter. A good example is the research paper titled Face Recognition Methods in Video Surveillance Systems Using Machine Learning [5]. It reviews different face identification models using deep learning techniques and describes how they can facilitate real-time security monitoring in video streams. In this type of system, the video is scanned to find faces using a convolutional neural network. When a face is detected, the model extracts identifying features and compares them with a database of authorized individuals. This enables the system to identify an individual without physical contact. These models can generate extremely realistic results when the lighting is good and faces are visible. Many face recognition methods perform poorly in low light or when a face is partly occluded, and most focus only on identifying people rather than vehicles or other objects that may also be important in security settings. The key limitation of this prior work is that it does not address intrusions involving vehicles. In our proposed system, face recognition is triggered only when someone enters a pre-defined restricted area. This keeps the analysis limited to relevant events and avoids processing data in situations that are not part of the security requirement.

- **Study 5: Automatic Number Plate Recognition (ANPR) Systems**

  Automatic Number Plate Recognition (ANPR) systems are commonly used to identify vehicles by extracting and reading license plate information from camera video. There is a research paper that illustrates this use of deep learning is "A Deep Learning-Based System for Automatic License Plate Recognition Using YOLOv12 and PaddleOCR." [6] In this study, they apply a recent YOLO-based detector to locate license plates and then run OCR to extract the characters. The system is designed to operate under typical real-world conditions, such as different lighting levels, camera angles, and moderate image blur. In ANPR systems like this, the process starts by locating the license plate using a deep learning object detector, such as YOLOv12. Once the plate area is detected, optical character recognition (OCR) tools such as PaddleOCR are used to read and extract the alphanumeric characters. After the characters are read, the license number can be compared with entries in a database to identify the vehicle. This approach supports fast and automated vehicle identification. OCR accuracy may decline when license plates are blurred, low resolution, or captured at steep angles. Most ANPR systems operate as stand-alone tools and are not integrated with person detection or restricted-area monitoring. A key limitation in earlier studies is that they do not connect their findings to the wider security context. This system addresses that limitation by combining ANPR with vehicle attribute recognition and tying both to authorization checks for vehicles entering a restricted area, so decisions are based on more than the license plate alone.

- **Research Gap and Proposed System**

The reviewed previous work shows that existing surveillance systems typically focus only on a single aspect of security, but do not provide complete protection from threats in multiple categories like human, animal and vehicle threat. Motion-based systems offten produce false alarms. Object detection systems lack identity verification. Face recognition systems ignore vehicle threats. ANPR systems operate independently without scene context. Therefore, the main research gap is the absence of a unified Return On Investment (ROI)-based surveillance system that provides these advantages like Ignore irrelevant public roadways, to detect and verify people, animal and vehicles, event-based alerts for people and vehicles. The Restricted Area Monitoring and Unauthorized Entry Detection System proposed in this paper addresses the above research gap by providing a unified framework for the development of a ROI-based surveillance system, which integrates ROI-based filtering, multi-class object detection, the provision of identity verification and face recognition, the provision of ANPR services, and the provision of event-based reporting to significantly reduce false alerts and increase the reliability of residential security systems.

Back to Table of Contents

4. Methodology

The proposed system of Restricted Area Monitoring and Unauthorize Entry Detection System has been divided into five major processing stages. It is clear that a structured and efficient process is followed in the proposed system. It is specified that the proposed system is using only videos fed from either a CCTV camera or an IP camera.

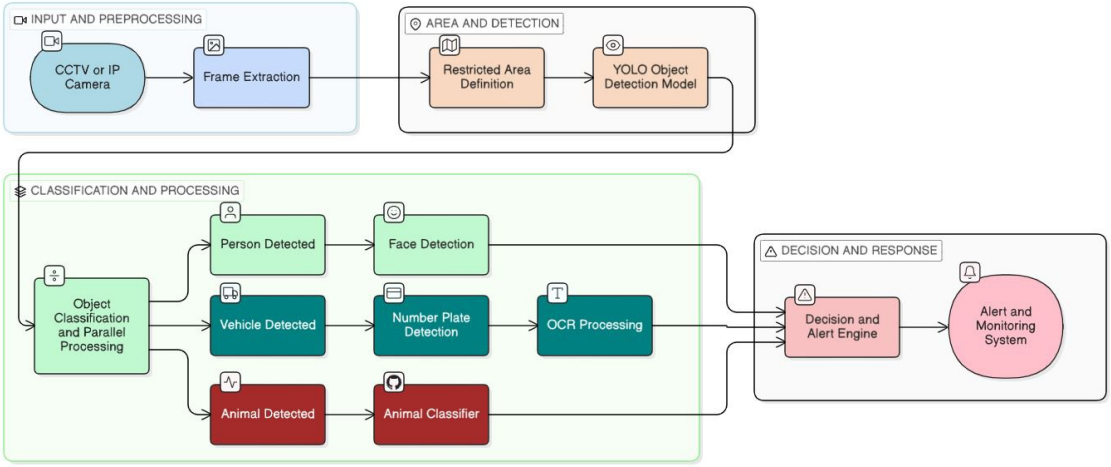


Figure 1: Workflow of Systeam

4.1. Input and Preprocessing

The Input and Preprocessing stage is the foundation of the proposed system, as the accuracy and reliability of all subsequent processing stages depend heavily on the quality of the input data. In this system, the only input source is a continuous video stream captured from CCTV or IP surveillance cameras installed in restricted areas.

4.1.1 Video Input Acquisition

The system constantly monitors the video footage being provided by fixed-surveillance cameras placed in a specific location that offers an optimal view of the restricted zone while attempting to minimize occlusions as much as possible. The data can be taken from real-time footage or from the recording of footage while the system is being tested or evaluated. Additionally, the system utilizes the video as the source of input in an attempt to offer a smoother service as well as utilizing changes over time such as movement or entries.

4.1.2 Frame Extraction

Since the computer vision technique as well as deep learning algorithms primarily process images rather than videos, the incoming stream of videos is split

into distinct images or frames, usually at a certain predefined rate (ranging from 15-30 frames per second). This process of obtaining images or data at distinct time points proves crucial since the computer can process every time instance individually without overlooking swiftly moving images like vehicles or animals.

4.1.3 Frame Resizing and Normalization

Each of the extracted frames is of a particular size suitable to the object detection model. Standardizing the frame sizes is appropriate to make the extracted frames compatible with the pre-trained deep networks while minimizing the complexity of the computations. Further, the application of different normalization methods to the pixels is necessary to standardize the pixel values to ensure the stability of the model.

4.1.4 Noise Reduction

Also, videos captured from the surveillance system contain noisy pixels caused mainly by low light intensity or other environmental effects on the captured image or the sensor used in the camera system. For this purpose, different image filtering methods such as Gaussian filtering can be used on the extracted image. Gaussian filtering reduces the noisy pixels in the image while maintaining the important visual features in the scene.

4.1.5 Noise Reduction

The surveillance footage is prone to noise due to low-light conditions, weather conditions or limitations on camera sensors. In order to overcome this challenge, noisy frames like Gaussian filtering or median filtering are used on the extracted frames. These methods minimize random noise and do not affect significant visual mechanisms including edges and contours of objects.

4.1.6 Lighting and Contrast Enhancement.

There are places that could be in daylight, nighttime, or artificial light conditions. For these reasons, brightness and contrast enhancement, such as histogram equalization or adaptive contrast adjustment, is applied for robust detection under all conditions. This enhances the visibility of objects and helps the detecting model to identify them correctly.

Once the preprocessing has been done the improved frames are sent to the restricted area detecting module. This will make sure that only standardized and good visual data is further analyzed to enhance the overall functionality of the system.

4.2. Restricted Area Detection

Restricted Area Detection stage involves detecting and tracking of certain areas in the field of view of the camera, where access should not be granted to the unauthorized individuals. This phase is very critical in the minimization of false detections and enhancing the computational efficiency.

4.2.1 Region of Interest (ROI) Definition

The restricted area is defined as a Region of Interest (ROI) which is manually defined through system set-up. ROI can be defined as rectangular or polygonal boundaries in order to fit into the shape of the constrained zone. This arrangement enables the system to be adapted to changing environments like construction sites, industrial premises or secure environments.

4.2.2 ROI Mask Generation

A binary mask to represent the ROI is made after the restricted area is defined. The pixels within the limited region are considered as valid and the pixels outside the area are not used in analysis. This is a masking technique that would make sure that only motions and objects within the defined restricted zone are taken into consideration hence a lot of unnecessary processing is avoided.

4.2.3 Object Presence Detection and Motion.

The system keeps checking the restricted zone on the presence or movement of objects. Simple motion detection algorithms, like frame differencing or background subtraction could be utilized to detect change in ROI. When there is no movement detected, the frame is thrown away thus saving on computational power. Once a movement or presence of an object is detected within the restricted area, such corresponding frames are sent to the object detection and classification phase to undergo a detailed analysis.

4.2.4 Rejection of irrelevant Movements

The slightest changes of shadows, light reflection, or background may cause false detections. Threshold based filtering and temporal coherence checks are used to counter this. The only movements that are to be considered as valid candidates to object detection are those that are sustained within a minimum period or meet some set threshold in size.

Without subjecting the object to movement, frames with valid movements of the object in the limited range are sent over to the object detection and classification

module. Frames that are not actively updated are discarded and efficient and focused processing is undertaken.

4.3. Object Detection and Classification

Object Detection and Classification stage is a main part of the suggested system, which recognizes and classifies objects that penetrate the restricted area. This step is used to make sure that every possible intrusion is properly detected and assessed by a human being, animal, or vehicles. In this step, an object detection model is selected and evaluated based on its detection performance and processing time.

4.3.1 Object Detection Model Selection

For achieving real-time performance and high detection accuracy, a deep learning-based object detection model is used. For this purpose, in this system, a YOLO-based object detection architecture is deployed due to its properties of being a one-stage detection model and successfully engaged in real-time Surveillance applications. This model has already been pre-trained on large datasets of different kinds of objects. It has then been fine-tuned for better detection accuracy.

4.3.2 Detection Process within Restricted Area

Every preprocessed frame containing motion within the restricted area is fed into the object detection model. The model analyzes it and detects objects by generating. Boxes bounding the entities detected. Class labels could be used to identify humans, animals, or vehicles. Confidence scores representing the reliability of each detection Only those detections that fall within the predefined restricted area and exceed a minimum confidence threshold are considered valid. This filtering step reduces false detections due to background elements or environmental noise.

4.3.3 Multi-Class Object Classification

Three categories of objects are detected and categorized:
Human
Vehicle
Animal
This multi-class approach to classification is necessary to cover a major shortcoming in existing security surveillance, where the sole objective is to detect humans. The detection approach also includes other entities like animals and vehicles, thereby increasing security coverage.

4.3.4 False Object and Noise Handling

Those which do not belong to the determined categories, or have low confidence levels, fall under the description of false or non-relevant objects. These false or non-relevant objects are discarded from the system. There is also a remarkable improvement in the efficiency of the system.

The output of this step contains classified object data with bounding box coordinates, object type, confidence scores, and frame timestamps. That information is being forwarded for further analysis to the specialized identification and authentication stage.

4.4. Specialized Identification and Authentication

The Specialized Identification and Authentication phase entails a comprehensive analysis on the identified objects based on the class to which the objects belong. This phase is responsible for improving the decision-making ability of the system through verification, additional information retrieval, and storage.

4.4.1 Human Detection and Face Analysis

If the detected object is identified as 'Human', then the subsequent process involves detecting the face present in that specific bounding box. A model is used that detects the facial features like eyes, nose, or mouth.

As mentioned in the problem, if a face is successfully detected:

(a) Facial regions are extracted and processed

(b) Identity verification process can be done by matching obtained feature details with an available database of authorized individuals (if available).

(c) Related information such as face images, detection timestamps, and location are stored in order to keep records

Even when the capability to verify user identities is not available, the detection of faces still increases the confidence of the presence of human figures and suppresses false human detection

4.4.2 Vehicle Detection and Number Plate Recognition

If the classification of the detected object is a vehicle, the system triggers a number plate detection process. To locate the region of the number plate, a vehicle bounding box is passed through a specialized license plate detection model.

Once the number plate is detected:

(d) Optical character recognition is done to identify and extract alphanumeric characters.

(e) The extracted number is compared against a database of authorized vehicles

The results of detection are then stored for further references, along with the respective vehicle's images and plate numbers. It is through this process that the system grants an effective distinction of vehicle entry between the authorized and unauthorized entries.

4.4.3 Animal Detection and Classification

Animals entering restricted areas may cause safety hazards or operational disruptions. Once an object has been classified as an animal, it is forwarded to an animal classification model that will assess the type of animal, such as a domestic or wild animal.

Detection data of animals, types, and timestamps would be recorded for monitoring and analysis.

4.4.4 Data Storage and Logging

For every type of object, information regarding the time of their detection, type, extracted features, and the frames captured is maintained in a database. This information helps in auditing, evaluating, and future improvements of the model.

The output of this stage includes authenticated or classified object information along with associated metadata. This information is passed to the decision-making module, which determines whether the detected entry is authorized or unauthorized.

4.5. Decision Making and Alert Generation

In this last stage, the information processed in the previous steps will be sent to the decision-making module. This module evaluates, with the predefined rules on object type, authentication status, and presence within the restricted area, if the detected entity is authorized or unauthorized. The system will produce an alert and log the event details along with the timestamp, object type, and captured frames when unauthorized entry is detected. In this way, these alerts will help security personnel take immediate action, thus improving overall monitoring efficiency.

Back to Table of Contents

5. Dataset / Data Collection

Data Collection

Accordingly, the restricted area detection system is to be trained and tested through combined publicly available data sets and self-collected data. The datasets have been carefully selected in order to cover the three main object classes of interest for the purposes of the project: Humans, Vehicles, and Animals.

1. Public Datasets

1. **Animal Detection**
    - **Dataset link:** animals-detection-images-dataset
    - The YOLO model is trained using this dataset to identify the presence of animals entering prohibited areas. It includes pictures of a variety of animal species taken in various settings, which helps the model learn accurate animal detection in practical settings.
2. **Vehicle Detection / Number Plate Recognition**
    - **Dataset:** vehicle-number-plate-detection
    - The YOLO model could be implemented to use this dataset to identify cars in restricted areas as well as license plates, especially when integrated with OCR, as it has pictures of cars or license plates that are annotated in its dataset.
3. **Human Detection**
    - **Dataset:** human-detection-dataset
    - The YOLO model that identifies the presence of humans in specific restricted areas is developed with the help of this dataset. This dataset contains images with people in different conditions, settings, with accurate annotations in the form of bounding boxes.

2. Self-Collected Data

The self-collected data is obtained from the actual restricted environment where the proposed system is likely to function. In this way, the model is evaluated under real-world conditions. Footage of the events is taken through the medium of surveillance cameras installed in the environment of interest.

- **Purpose and Description:**
    - The videos thus collected would also be used for evaluating the system's performance in real- scenarios which can arise in the world, such as the movement

of vehicles, intrusion of animals, or the presence nce within restricted areas, including environments with varying light- ties, camera angles, and occlusion.

  - Furthermore, the data will be utilized in testing the YOLO model together with second-stage, specialized detection models, such as face detection, animal classification, as well as vehicle number plate detection with OCR in the target environment.

- **Data Collection Method:**

  - Short clips are used that resemble human or animal functionality movement, and vehicle activity in controlled restricted zones.
  - The videos that are recorded are processed by obtaining the different frames in what can be described as – and then used for the testing of models and evaluating performance..

3. Data Preparation for YOLO

- All the public and self-collected data will be converted to use the common format for data annotation with YOLO.
- The object classes of the first-stage YOLO model can be represented as:
  1. Person → Human
  2. Vehicle → Vehicle
  3. Animal → Animal
- Such a unified labeling process enables the detection of humans, vehicles, as well as animals, by the YOLO model with efficiency, thus reducing the complexity associated with the monitoring process in restricted areas in real-time.

6. Expected Outcomes

The need for an efficient, accurate, and automated facility for the monitoring process will be met by the proposed Restricted Area Monitoring and Unauthorized Entry Detection System. It is expected that computer vision and image processing will help to overcome the problems of the current state-of-the-art for surveillance systems with the aid of human monitoring.

One of the first expected outcomes is the identification of any unauthorized entry in specific defined areas of restricted movement. The system is expected to be effective in detecting humans, animals, and even cars individually. Thus, there is no way security can be compromised since all possibilities will be addressed. This is an improvement from other systems that can only detect human movement.

Another notable outcome of the proposed approach is the elimination of false alarms. By limiting the fields of concern only to the designated regions of interest, or ROI, the use of preprocessing techniques helps ensure the elimination of false alarms due to the presence of shadows or environmental conditions. This helps ensure increased reliability.

Moreover, a system for specific identification and authentication will also be supported. Human intrusions can also be identified and analyzed further through face detection and recognition, while vehicular intrusions will be identified through number plate recognition. Thus, the system will have greater ability to distinguish between authorized and unauthorized entities.

Another key outcome is the automation of surveillance operations. By reducing reliance on manual CCTV monitoring, the system decreases human workload, minimizes human error, and improves consistency in surveillance performance. Stored detection data can also be used for system evaluation, security analysis, and future model improvements.

Another major outcome is that the surveillance operations become computerized. This reduces the involvement of humans, thereby decreasing the workload, preventing any chances of error, and ensuring that consistency is maintained in the overall efficiency of the surveillance posture that is adopted. The stored detection data can also be employed to evaluate the system, security, etc.

Generally, the result will be the establishment of an effective, expandable, and wise surveillance system to enrich security, monitor events more effectively, and make decisions wisely in constrained areas like industrial areas, construction sites, etc.

Back to Table of Contents

7. Percentage of Image Processing & Computer Vision Involvement

This project is heavily dependent upon Image Processing and Computer Vision domains, as all functionalities of this system are based upon an analysis of processed and processed visuals. Estimate of total involvement of Image Processing and Computer Vision domains:

1. **Image Processing: Approximately 40%** Image processing algorithms are greatly used in the preprocessing steps of the videos, which are concerned with image resizing, normalization, reduction of image noises, image brightening, image contrast improvement, detection of image movement, detection of the region of interest, etc.
2. **Computer Vision: Approximately 50%** The major components are computer vision, which comprises object detection, object classification, face detection, animal detection, vehicle detection, and number plate recognition. The system decision mechanisms are based greatly on deep learning with regard to vision and extraction of visual features; therefore, computer vision is the major component.
3. **Other Components (Decision Logic, Alerts, Data Handling): Approximately 10%** This encompasses rules-based authorization, alerts, storage, logging, as well as system controls.

Overall Total Image Processing Computer Vision Contribution: 90%

Back to Table of Contents

References

[1] D. K. Yadav, L. Sharma, and S. K. Bharti, "Moving object detection in real-time visual surveillance using background subtraction technique," in *2015 International Conference on Computing, Communication and Automation (ICCCA)*, 2015.

[2] P. K. Macherla, A. Telagareddi, N. Kollipara, and H. B. Bommareddy, "Real time moving object detection using yolo," *International Journal for Research in Applied Science & Engineering Technology (IJRASET)*, 2025.

[3] D. J. M. Reddy, M. H. Vardhan, M. K. Nadh, and N. Priyanka, "Real time object detection based on yolov3 using python," *International Journal of Engineering Research & Technology (IJERT)*, 2023.

[4] H.-G. Jeon and K.-H. Lee, "Region-of-interest extraction method to increase object-detection performance in remote monitoring system," *Applied Sciences*, 2025.

[5] F. P. Mahdi, M. M. Habib, M. A. R. Ahad, S. Mckeever, A. Moslehuddin, and P. Vasant, "Face recognition-based real-time system for surveillance," *Intelligent Decision Technologies*, 2017.

[6] B. Buleu, R. Robu, and I. Filip, "A deep learning-based system for automatic license plate recognition using yolov12 and paddleocr," *Applied Sciences*, 2025.

[1] [2] [3] [4] [5] [6]